

Question Paper

Exam Date & Time: 22-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Pattern recognition [IT_4403]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze how the Bayesian decision rule minimizes the probability of classification error when prior probabilities are unequal. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Analyze how continuous and discrete feature distributions influence decision boundaries in Minimum-Error-Rate classification. (5)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Apply Maximum Likelihood Estimation to determine the parameters of a Gaussian distribution for a given small dataset and verify the fit graphically. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Explain the K-Nearest Neighbor (KNN) algorithm and analyze how the choice of 'K' affects classification accuracy. (5)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Explain why non-parametric classification methods such as k-NN or Parzen Window classifiers are more suitable when the underlying class distribution is unknown or non-linear. (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Explain how Support Vector Machines (SVM) find the optimal hyperplane for classification. Derive the mathematical formulation for maximizing the margin between two classes and discuss the role of support vectors in determining the decision boundary. (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Apply the Perceptron learning rule to update the weight vector for one training iteration given specific feature inputs, targets, and learning rate. (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Application"
- B) A marketing dataset contains information on whether customers purchase a product based on income level. The data distribution is as follows:

Income Level	Yes	No	Maybe	Total
High	3	1	0	4
Medium	2	2	1	5
Low	1	3	1	5

1. Calculate the Entropy for each income level.
2. Find the Weighted Average Entropy.
3. Determine the Information Gain if the overall dataset has 10 samples (Yes=6, No=4, Maybe=1).

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Perform Single Linkage Hierarchical Clustering step by step using the following Euclidean distance matrix. (5)
- A) At each iteration, identify which clusters merge, show the updated distance matrices, and draw the dendrogram representing the cluster formation.

	1	2	3	4	5
1	-	4.0	11.7	20.0	21.5
2	4.0	-	8.1	16.0	17.9
3	11.7	8.1	-	9.8	9.8
4	20.0	16.0	9.8	-	8.0
5	21.5	17.9	9.8	8.0	-

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Explain how the DBSCAN algorithm identifies core, border, and noise points using the parameters ϵ (*epsilon*) and *MinPts*. (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4403:Pattern recognition

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 IT_4403:Pattern recognition

2026

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	50(100%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)